# Connery J. Chen

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## Education

**University of Nevada, Las Vegas** — Las Vegas, NV
Ph.D. in Astronomy | Advisor: Bing Zhang — 2021–May 2026
Relativistic Transients from Compact Objects: Radiation Processes and Geometric Effects

**Santa Clara University** — Santa Clara, CA
B.S. in Physics | Minor in Mathematics — 2019–2021

**Hack Reactor** — San Francisco, CA
Software Engineering Immersive — 2017–2018

## Research Experience

**Louisiana State University** — Baton Rouge, LA
Postdoctoral Researcher — 2026–Current

- High-energy, time-domain, and multi-messenger astrophysics; scientific software development
- Developing integrated transient discovery and classification pipelines
- Conducting interdisciplinary research in AI, medical physics, and astrophysics

**NASA Goddard Space Flight Center** — Greenbelt, MD
Student Researcher — 2025–2026

- Theoretical high-energy time domain and multi-messenger astrophysics
- Led independent research projects resulting in one manuscript published in ApJ
- Presented research at international conferences and scientific meetings

**University of Nevada, Las Vegas** — Las Vegas, NV
Graduate Assistant — 2021–2026

- Theoretical and computational high-energy and multi-messenger astrophysics
- Published 3 first-author papers in MNRAS, ApJ and ApJL
- Managed and coordinated regular outreach events in the Las Vegas community
- Mentored undergraduate research projects for two students

**SOFIA Science Center—NASA Ames Research Center** — Mountain View, CA
Research Intern — 2020–2021

- Polarization of the Interstellar Medium | Advisors: Dr. B-G Andersson and Dr. Kristin Kulas
- Provided novel constraints on grain growth contributions to ISM polarization
- Presented research at USRA, Santa Clara University, and AAS 238

## Fellowships and Awards

- Nevada NASA Space Grant Graduate Research Opportunity Fellowship — Fall 2024–Spring 2025
- UNLV Top Tier Doctoral Graduate Research Assistantship — Fall 2024–Spring 2026

## Publications

**Chen**, **C. J.**, Wang, Y., and Zhang, B. (Aug. 2026). "Einstein Probe Fast X-Ray Transients Extend the Physical Parameter Space of Relativistic Jets". In: *ApJL* 1006.2, L44. DOI: 10.3847/2041-8213/ae89a4.

**Chen**, **C. J.**, Wang, Y., and Zhang, B. (Apr. 2026). "X-Ray Emission Signatures of Neutron Star Mergers". In: *ApJ* 1001.2, 177. DOI: 10.3847/1538-4357/ae5535.

Wang, Y., **Chen**, **C. J.**, and Zhang, B. (Feb. 2026). "VegasAfterglow: A high-performance framework for gamma-ray burst afterglows". In: *JHEAP* 50, 100490. DOI: 10.1016/j.jheap.2025.100490.

**Chen**, **C. J.** and Zhang, B. (Mar. 2023). "FRB-SRB-XRB: Geometric and relativistic beaming constraints of fast radio bursts from the Galactic magnetar SGR J1935+2154". In: *MNRAS* 519.4. DOI: 10.1093/mnras/stac3747.

## Selected Talks and Presentations

- **Invited Talk**: Gamma-Ray Speakeasy | NASA GSFC — Sept. 2026
  *X-Ray Emission Signatures from Neutron Star Mergers*
- **Contributed Talk**: 4th Multi-Messenger Symposium | Nevada Center for Astrophysics — Apr. 2026
  *Einstein Probe Fast X-Ray Transients Highlight Tension with Standard Off-Axis GRBs*
- **Invited Talk**: Astronomy Seminar | George Washington University — Nov. 2025
  *X-Ray Emission Signatures from Neutron Star Mergers*
- **Poster**: 4th Time Domain and Multi-Messenger Astrophysics Workshop | NASA MSFC — Oct. 2025
  *On the Jet Structure of Fast X-Ray Transients*
- **Poster**: High-Energy Astrophysics Division Conference | American Astronomical Society — Oct. 2025
  *X-Ray Emission Signatures from Neutron Star Mergers*
- **Contributed Talk**: 3rd Multi-Messenger Symposium | Nevada Center for Astrophysics — Feb. 2025
  *X-Ray Emission Signatures from Neutron Star Mergers*
- **Poster**: GRB50: The Past, Present, and Future of Gamma-Ray Bursts — Aug. 2023
  *Numerical Modeling of TeV Photon Emission from GRB 221009A*
- **Invited Talk**: Symposium on Fast Radio Bursts and Their Astrophysics — May 2023
  *FRB-SRB-XRB: Geometric and Relativistic Beaming Constraints of Fast Radio Bursts from the Galactic Magnetar SGR J1935+2154*
- **Poster**: 1st Multi-Messenger Symposium | Nevada Center for Astrophysics — Feb. 2022
  *FRB-SRB-XRB: Geometric and Relativistic Beaming Constraints of Fast Radio Bursts from the Galactic Magnetar SGR J1935+2154*
- **Poster**: 238th American Astronomical Society Meeting — Jun. 2021
  *Radiative Alignment Torque Modeling*
- **Seminar**: Universities Space Research Association Annual Symposium — Sept. 2020
  *Radiative Alignment Torque Modeling*

## Skills

- **Software Development:** Python, C, C++, JavaScript, MATLAB, Git, Linux/Unix
- **Computational Science:** data analysis, machine learning, numerical simulations, statistical modeling, MCMC, parallel computing, HPC
- **Scientific Communication:** scientific visualization, technical writing, public speaking and outreach

## Scientific Software

See full list of projects on https://github.com/cjules0609/

- VegasAfterglow (C++, Python)
  High-performance numerical framework for modeling GRB afterglows.
- PromptX (Python)
  Numerical framework for modeling prompt emission from relativistic outflows.

## Teaching

- **Laboratory Instructor** | The University of Nevada, Las Vegas — Spring 2022–Fall 2024
  *Introduction to Astronomy (AST 105) & General Physics I (PHYS 105)*
- **Laboratory Teaching Assistant** | The University of Nevada, Las Vegas — Fall 2021
  *Introduction to Astronomy (AST 105)*
- **Laboratory Teaching Assistant** | Santa Clara University — Spring 2020
  *Physics for Scientists and Engineers IV (PHYS 34)*

## Additional Experience

**Proxima Capital, LLC** — Santa Clara, CA
Co-founder and CTO — 2016–2019

- Co-founded a quantitative trading firm specializing in Foreign Exchange markets
- Developed algorithmic strategies and automated monitoring systems for real-time market analysis
- Built integrated MQL4 and Python pipelines for data ingestion, processing, and storage

**Age of Learning Inc.** — Glendale, CA
Assistant Product Manager — 2015–2016

- Software engineer for the product management team
- Developed a comprehensive employee training course using fullstack technologies
- Built a fullstack interactive floor plan to support internal operations